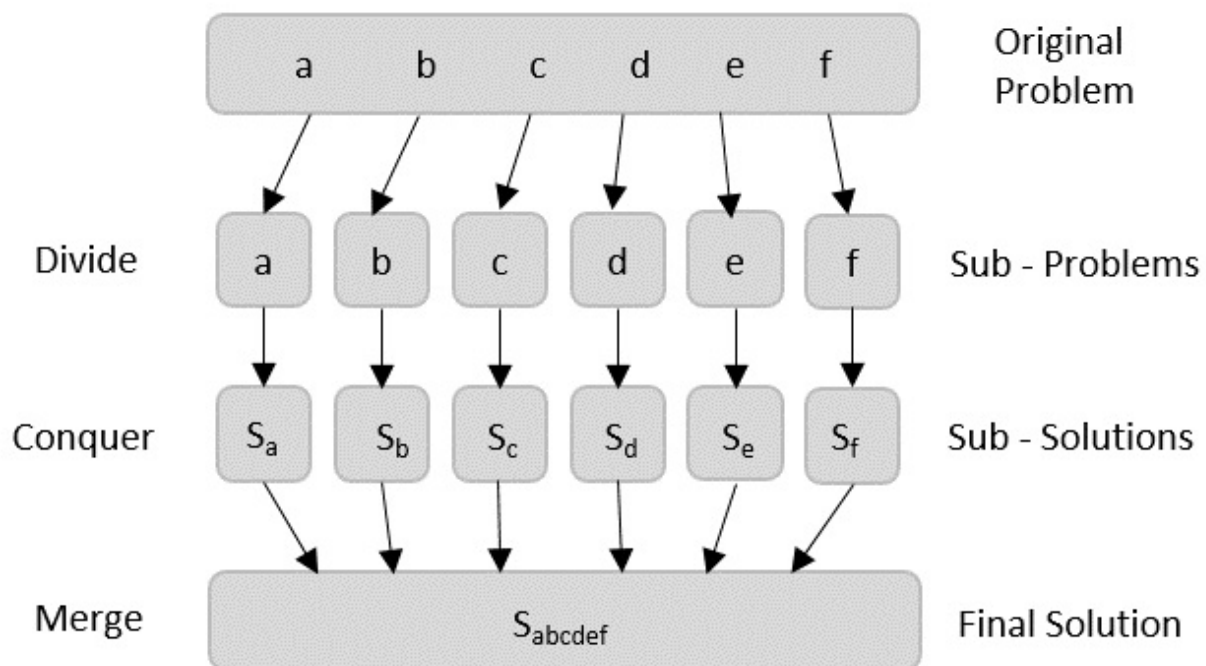


Design and Analysis - Divide and Conquer

Using divide and conquer approach, the problem in hand, is divided into smaller sub-problems and then each problem is solved independently. When we keep dividing the sub-problems into even smaller sub-problems, we may eventually reach a stage where no more division is possible. Those smallest possible sub-problems are solved using original solution because it takes lesser time to compute. The solution of all sub-problems is finally merged in order to obtain the solution of the original problem.



Broadly, we can understand **divide-and-conquer** approach in a three-step process.

Divide/Break

This step involves breaking the problem into smaller sub-problems. Sub-problems should represent a part of the original problem. This step generally takes a recursive approach to divide the problem until no sub-problem is further divisible. At this stage, sub-problems become atomic in size but still represent some part of the actual problem.

Conquer/Solve

This step receives a lot of smaller sub-problems to be solved. Generally, at this level, the problems are considered 'solved' on their own.

Merge/Combine

When the smaller sub-problems are solved, this stage recursively combines them until they formulate a solution of the original problem. This algorithmic approach works recursively and conquer & merge steps work so close that they appear as one.

Pros and cons of Divide and Conquer Approach

Divide and conquer approach supports parallelism as sub-problems are independent. Hence, an algorithm, which is designed using this technique, can run on the multiprocessor system or in different machines simultaneously.

In this approach, most of the algorithms are designed using recursion, hence memory management is very high. For recursive function stack is used, where function state needs to be stored.

Examples of Divide and Conquer Approach

The following computer algorithms are based on divide-and-conquer programming approach –

- Merge Sort
- Quick Sort
- Binary Search
- Strassen's Matrix Multiplication
- Closest pair (points)
- Karatsuba

There are various ways available to solve any computer problem, but the mentioned are a good example of divide and conquer approach.
